

The need for WebRTC in the enterprise

Open source technology from Google, which is now a standard in development with the IETF (Internet Engineering Task Force), could lead to inexpensive and interoperable video solutions in the business world, where large enterprises have been forced to invest hundreds of thousands of dollars in collaboration architecture, and smaller companies couldn't even consider the technology due to price and complexity.

Sending data between two users in today's browser world is a tough process with JavaScript, and most developers rely on a server as the middle agent. With WebRTC, the Data Channel API allows the passing of arbitrary data across the connection. This allows for large scale file sharing, fast action multiplayer games, and even remote control applications. The Data Channel allows users to transfer large amounts of data very quickly, all in the Web browser in true peer-to-peer mode.

WebRTC adds advanced levels of interaction to a website, allowing users to communicate in-context and in real-time, with either site operators or with each other. Such capabilities overcome current limitations that require users to call a number, download a plugin, or leave the website.

The uses of WebRTC

WebRTC use-cases can range from basic video/voice chat, to multiplayer gaming, file sharing and so on. Listed below are some of the possibilities.

Video recruitment and interviewing

Video interviews and recruiting have become more popular as enterprises and small businesses try to make the hiring process more efficient. Companies could conduct video interviews just by sharing a URL with the candidate and thereby decrease the cost of hiring. They could screen candidates faster, and the video session could be recorded and shared for further review.

Live video technical app troubleshooting support

Video customer support for Web or mobile applications gives customers a better user experience. The customer support executive could interact and troubleshoot issues with users on their device. By using WebRTC's screen sharing, customer service agents could save time while troubleshooting with customers. Users could grant real-time device control to the support team to troubleshoot and eliminate confusing communication barriers.

Field service video calling

In the field, service technicians could use the camera on smartphones or tablets to contact experts located at a remote location to get help on troubleshooting devices and equipment, eliminating the need to send an expert to a remote location. This would reduce service downtime, decrease

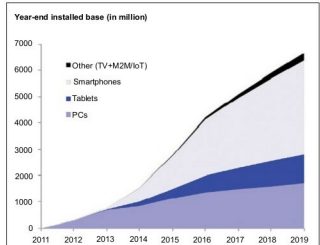


Figure 3: WebRTC market forecast

operational costs and improve customer satisfaction.

Online shopping assistants and concierge services

Online retail shopping assistants and concierge services could use video calls to increase brand loyalty, sales and customer satisfaction. When customers seek advice on an outfit or an accessory for their outfit or when an agent is consulted on the use of makeup to complement the client's features, there is increased customer satisfaction. The common occurrence of abandoning the shopping cart can be reduced by connecting with customers in real-time and thereby increasing sales as well as brand loyalty.

Equipment or product troubleshooting

Customers could use a video to show a customer service agent issues with a specific item and configuration before sending or returning the product. This could save time while troubleshooting with customers, and also reduce processing and shipping costs for returning the product.

Market trends

Since the initial release of the WebRTC API by Google in May 2011, the project has grown both in terms of support and interest. According to Dean Bubley's 'Disruptive Analysis', nearly three billion devices are WebRTC-capable today, with an estimate of about 6.5 billion devices supporting WebRTC by 2019. Today, desktop browsers lead in WebRTC usage, with Google Chrome, Mozilla Firefox and Opera accounting for more than 60 per cent of total browser usage. In the years to come, usage will increase quickly with the availability of browser support on smartphones.

By 2018, there will be an interesting traction for WebRTC within the Internet of Things (IoT) space. By definition, IoT connects things to the Web to interact with them in real-time. So, it makes sense that IoT, combined with WebRTC, can create a promising future with real-time communications.